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(EGYPTIAN / 25 YEARS)

[View All Certificates\(27+\)](#)

Junior Data Scientist

(Open to Internship Opportunities)

SUMMARY

- Junior Data Scientist specializing in building end-to-end machine learning solutions, from raw data to deployed, production-ready models. Skilled in predictive maintenance, time-series forecasting, and classification problems across industries including healthcare, aviation, and energy. Delivered 13 hands-on projects covering feature engineering, model evaluation, and real-world deployment. Strong foundation in Python, SQL, and statistical analysis, backed by 27+ certifications spanning Data Science fundamentals, Statistics, and Machine Learning. Focused on turning complex, messy data into clear, actionable insights that support real business decisions.

SKILLS

- Python & SQL for data analysis and querying
- Pandas & NumPy for data manipulation and numerical computing
- Statistical Analysis, EDA & Hypothesis Testing
- Machine Learning (Regression, Classification, Clustering)
- Feature Engineering & Data Preprocessing
- Model Evaluation, Cross-Validation & Performance Metrics
- Time Series Forecasting & Trend Analysis
- Data Visualization (Matplotlib, Seaborn)
- Predictive Analytics & Data-Driven Decision Making
- Data Cleaning, Handling Missing Values & Outliers
- Analytical & Critical Thinking
- Business-Oriented Data Interpretation

CERTIFICATES

IBM Data Science Professional Certificate | Coursera

IBM Machine Learning | Coursera

Data Analysis with Python | Coursera

Deep Learning & Reinforcement Learning | Coursera

SQL Certification | DataCamp

Explore AI & Generative AI | Microsoft Learn

Certification Generative AI Elevate Your Data Science Career (Coursera)

Certification Machine Learning with Python (Coursera)

PROJECTS

Lithology Classification from Well Logs

- Built and deployed an end-to-end machine learning system classifying subsurface rock lithology from well log measurements across 1.17M+ readings and 108 wells, using a Random Forest model trained on FORCE 2020 and Kansas SEG 2016 public datasets.
- Diagnosed and resolved a critical data quality issue where 99.7% of the merged dataset lacked target labels, recovering full labeling by re-matching against the original source data on well and depth (100% match rate on 1.17M rows).
- Identified and cleaned physically invalid sensor readings (e.g., Photoelectric Factor values 25x beyond real-world range) across the full dataset, improving data integrity before model training.
- Achieved 66% accuracy and 0.33 macro-F1 across 20 imbalanced lithology classes, validated using a well-level train/test split to prevent data leakage and ensure realistic generalization to unseen wells.
- Deployed a production-ready web application (Streamlit, hosted on Streamlit Cloud with Hugging Face Hub model serving) allowing real-time batch and single-point lithology predictions with confidence scoring for geological review prioritization.

Predictive Maintenance for Medical Equipment

- Engineered a predictive maintenance framework using Gradient Boosting and Random Forest to estimate the Remaining Useful Life (RUL) of critical medical assets, achieving an ($R^2 \approx 0.30$) and a Mean Absolute Error of 76 days.
- Developed complex interaction features, such as Age $\times$ Usage and Temp $\times$ Volt, to capture non-linear failure triggers, increasing model sensitivity to operational stress beyond chronological age.
- Implemented Hierarchical Clustering to group 7,000+ medical devices (e.g., Ventilators, MRIs) by failure signatures, identifying that maintenance history accounts for up to 46.8% of total failure risk.
- Mitigated data leakage by systematically isolating target-correlated variables, ensuring a robust and unbiased model capable of generalizing across diverse hospital sensor data.
- Extracted actionable business insights that enable a transition from reactive to proactive maintenance, potentially reducing hospital equipment downtime and optimizing CAPEX allocation.

Predictive Maintenance for Aircraft Engines

- Built an end-to-end predictive maintenance system for aircraft engines using the NASA C-MAPSS time-series dataset with multivariate sensor data.
- Reframed Remaining Useful Life (RUL) prediction into a 4-class maintenance classification problem (Healthy, Soon, Warning, Critical) to support operational decision-making.
- Prevented data leakage by performing engine-level train/validation splits, ensuring realistic evaluation on unseen engines.
- Engineered temporal features (rolling mean, rolling standard deviation, and cycle-to-cycle deltas), expanding the feature set from 24 to 87 features.
- Trained a Random Forest classifier achieving ~85% validation accuracy, with ~97% recall for Healthy cases and ~88% recall for Critical failures.
- Enhanced model interpretability by analyzing top sensor importance, highlighting key warning signals used to identify imminent engine failures.

Inview: AI-Driven Bilingual Video Interviewing & Proctoring Platform

- Developed an end-to-end AI platform for automated candidate screening using LLMs (Qwen 2.5) and Speech-to-Text (Whisper-v3) for deep semantic analysis of interviews.
- Engineered a Bilingual NLP Pipeline (Arabic/English) capable of extracting technical proficiency, communication scores, and behavioral traits from unstructured audio data.
- Implemented a Computer Vision (OpenCV) proctoring system to monitor candidate presence and detect multi-face security alerts during live sessions.
- Built a professional-grade Glassmorphism Dashboard using Streamlit, featuring dynamic AI-generated reports and a multi-factor scoring matrix for HR decision-making.
- Automated the deployment pipeline to Streamlit Cloud, integrating complex data workflows with real-time video processing.